

Estimating Peer Effects in School Networks: A Replication Note of Boucher et al. (2026)

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This note describes the model and estimation procedure used in Boucher et al. (2026). The goal is to provide a clear and concise summary of the key components of the model, the econometric challenges, and the estimation strategy. The note is designed to be self-contained and consistent with the Python code used for replication in class.

The logic of the replication proceeds from simulating the environment to estimating the model. The first step is to generate a social network of students within schools. The second step is to simulate outcomes based on the model's assumptions. The third step is to estimate the model parameters using GMM. The advantage of this approach is that it allows for a clear understanding of the model's structure and the estimation procedure, providing full control on the data generating process. We focus on GPA as the outcome of interest, but the model can be applied to other outcomes as well.

1 School Networks

There are N students distributed across S schools. Students interact only with other students in the same school. Let A denote the friendship adjacency matrix. The entry $a_{ij} = 1$ if student i names student j as a friend, and $a_{ij} = 0$ otherwise. The network is directed. Therefore, it is possible that $a_{ij} = 1$ while $a_{ji} = 0$. In other words, student i may name student j as a friend even if student j does not name student i . We construct the weighted adjacency matrix G by row-normalizing A . If student i names at least one friend, then $g_{ij} = \frac{a_{ij}}{\sum_k a_{ik}}$. If student i names no friends, then student i is isolated and the entire row of G is set equal to zero. There are no friendship links across schools. Let N_N denote the number of non-isolated students, and let $N_I = N - N_N$ denote the number of isolated students.

2 Model

Let y_i denote the GPA chosen by individual i , and let g_{ij} denote the weight that individual i assigns to peer j . For non-isolated individuals, the social norm faced by individual i is a CES aggregator of peers' outcomes:

$$\tilde{y}_i(\beta, \mathbf{y}) = \tilde{y}_i(\beta) = \left(\sum_j g_{ij} y_j^\beta \right)^{1/\beta}.$$

The parameter β governs the shape of the social norm. When $\beta = 1$, the social norm is the weighted average of peers' outcomes. As β becomes large, the norm places more weight on high-outcome peers. In the limit as $\beta \rightarrow \infty$, the norm becomes the maximum peer outcome.

The utility function (for non-isolated individuals) is

$$U_i(y_i, \mathbf{y}, \mathbf{g}) = \alpha_i y_i + \theta_1 y_i \tilde{y}_i - \frac{1}{2} \left[y_i^2 + \theta_2 (y_i - \tilde{y}_i)^2 \right].$$

The parameter θ_1 captures spillovers: the marginal benefit of choosing a higher y_i increases with the social norm $\tilde{y}_i$. The parameter θ_2 captures conformism: the individual dislikes choosing an outcome far from the social norm.

The first-order condition with respect to y_i can be written as

$$y_i = \frac{1}{1 + \theta_2} \alpha_i + \frac{\theta_1 + \theta_2}{1 + \theta_2} \tilde{y}_i.$$

Let $\alpha_i = x_i' \gamma + \varepsilon_i$, where x_i is a vector of observable covariates, and define $\lambda_2 = \frac{\theta_2}{1 + \theta_2}$, $\lambda_1 = \frac{\theta_1}{1 + \theta_2}$.

Then, for non-isolated individuals, the equilibrium choice equation can be written as

$$y_i = (1 - \lambda_2) (x_i' \gamma + \varepsilon_i) + (\lambda_1 + \lambda_2) \tilde{y}_i(\beta).$$

3 Relabeling for Estimation

This section introduces a relabeling that is useful for estimation and coding. Define

$$\delta = 1 - \lambda_2 = \frac{1}{1 + \theta_2}, \quad \lambda = \lambda_1 + \lambda_2 = \frac{\theta_1 + \theta_2}{1 + \theta_2}, \quad e_i = \delta \varepsilon_i.$$

Then the non-isolated outcome equation can be written as

$$y_i = \delta x_i' \gamma + \lambda \tilde{y}_i(\beta) + e_i.$$

For isolated individuals, the social norm is absent. Their outcome is therefore determined only by their own covariates and unobserved component:

$$y_i = x_i' \gamma + \varepsilon_i.$$

4 Econometric Challenges

There are two main challenges for estimation. First, the social norm $\tilde{y}_i(\beta)$ depends on β in a nonlinear way. Second, the model includes an endogenous regressor. The peer norm $\tilde{y}_i(\beta)$ is constructed using peers' actual outcomes, and these peer outcomes are by construction a function of individual i 's outcome. We follow the paper abstracting away from two potential concerns by considering the network as exogenous and assuming correlated effects to be captured by school fixed effects.

5 Identification Strategy and Instruments

The estimation relies on exclusion restrictions, which are discussed below and motivate a GMM estimator to recover the model parameters. The basic idea is to use instruments based on peers' covariates to isolate exogenous variation in peers' outcomes.

Let $\hat{\mathbf{y}}$ be the vector of predicted outcomes obtained as a linear function of covariates. A potentially relevant instrument for peer social norm $\tilde{y}_i(\beta)$ is the social norm emerging from the component of their outcome predicted by their observable covariates, defined as $\tilde{y}_i(\beta, \hat{\mathbf{y}})$

$$\tilde{y}_i(\beta, \hat{\mathbf{y}}) = \left(\sum_j g_{ij} \hat{y}_j^\beta \right)^{1/\beta}.$$

Furthermore, the derivative $\tilde{y}'_i(\beta, \hat{\mathbf{y}})$ captures how the predicted social norm changes as the curvature parameter β changes.

The vector of instruments for non-isolated individuals is

$$z_i(\beta) = (x'_i, \tilde{y}_i(\beta, \hat{\mathbf{y}}), \tilde{y}'_i(\beta, \hat{\mathbf{y}}))'.$$

Thus, the instruments include:

- (i) individual covariates x_i ;
- (ii) the social norm of peers' predicted outcomes, $\tilde{y}_i(\beta, \hat{\mathbf{y}})$;
- (iii) the derivative of the predicted social norm with respect to β , $\tilde{y}'_i(\beta, \hat{\mathbf{y}})$.

The estimator will recover the parameters $(\gamma, \lambda, \beta, \delta)$ using GMM, under the assumption that the following moment conditions hold for isolated and non-isolated individuals, respectively:

$$E[x_i \varepsilon_i] = 0 \quad E[z_i(\beta) \varepsilon_i] = 0.$$

6 GMM Estimator

For isolated individuals, the moment condition is $E[x_i \varepsilon_i] = 0$.

The corresponding $(k \times 1)$ vector of sample moments is:

$$m_I(\gamma) = \frac{1}{N_I} \sum_{i \in I} x_i (y_i - x_i' \gamma).$$

For non-isolated individuals, the moment condition is $E[z_i(\beta) e_i] = 0$.

The corresponding $(k + 2 \times 1)$ vector of sample moments is:

$$m_N(\gamma, \lambda, \beta, \delta) = \frac{1}{N_N} \sum_{i \in N} z_i(\beta) (y_i - \delta x_i' \gamma - \lambda \tilde{y}_i(\beta, \mathbf{y})).$$

The GMM objective function is

$$Q(\gamma, \lambda, \beta, \delta) = m_I(\gamma)' W_I m_I(\gamma) + m_N(\gamma, \lambda, \beta, \delta)' W_N m_N(\gamma, \lambda, \beta, \delta),$$

where W_I and W_N are weighting matrices. The GMM estimator chooses the parameter vector that minimizes deviations of the sample moments from zero:

$$(\hat{\gamma}, \hat{\lambda}, \hat{\beta}, \hat{\delta}) = \arg \min_{\gamma, \lambda, \beta, \delta} Q(\gamma, \lambda, \beta, \delta).$$

In general, this problem does not have a closed-form solution because the objective function is non-linear in β . Therefore, the estimator searches over the parameter space and chooses the parameter vector that minimizes the GMM objective.

7 Numerical Estimation

We implement the following concentrated two-step GMM procedure.

Preliminary Step

1. Compute predicted outcomes $\hat{\mathbf{y}}$.
2. Choose starting values for the parameter vector (λ, β, δ) .
3. Choose initial weighting matrices: identity matrices.

First-Step GMM

For each candidate parameter vector:

1. Construct the social norm $\tilde{y}_i(\beta, \mathbf{y})$ using actual peers' outcomes.
2. Construct the instrument $\tilde{y}_i(\beta, \hat{\mathbf{y}})$ using predicted peers' outcomes.
3. Construct the derivative $\tilde{y}'_i(\beta, \hat{\mathbf{y}})$.
4. Compute $\hat{\gamma}(\lambda, \beta, \delta)$ by solving the linear GMM problem conditional on (λ, β, δ) .
5. Compute the residuals for isolated and non-isolated individuals (subtract school fixed effects).
6. Construct the sample moments $m_I(\gamma)$ and $m_N(\gamma, \lambda, \beta, \delta)$.
7. Evaluate the GMM objective function.
8. Update the parameter vector according to the optimization algorithm.

These steps are repeated until the GMM objective function cannot be reduced further.

Second-Step GMM

After obtaining the first-step estimates:

1. Compute the residuals and moments at the first-step estimates.
2. Estimate the covariance matrices of sample moments for isolated and non-isolated individuals.
3. Set the second-step weighting matrices equal to the inverse of the estimated covariance matrices.
4. Repeat the first-step GMM procedure using the updated weighting matrices.

The resulting estimates are the two-step GMM estimates.

8 Why Concentrate the GMM Problem?

The direct procedure is conceptually simple: search over all parameters, compute residuals, compute moments, and choose the parameter vector that makes the moments as close to zero as possible.

However, each additional parameter increases the dimension of the numerical search problem. In practice, this can make optimization slower and less stable because the optimizer must explore many possible parameter combinations. Avoiding the numerical search over even one block of parameters can therefore bring substantial computational gains.

The key observation is that one block of parameters is special. Once we fix a candidate value of the nonlinear parameters (λ, β, δ) , the coefficient vector γ enters the model linearly. Linear-GMM algebra can then be used to solve for the best value of γ conditional on (λ, β, δ) .

Therefore, the GMM problem can be concentrated. Instead of searching over the full parameter vector $(\gamma, \lambda, \beta, \delta)$, the algorithm searches only over (λ, β, δ) . For every candidate value of (λ, β, δ) , it solves a nested linear problem for γ .

To see this, fix a candidate value of (λ, β, δ) . Then $\tilde{y}_i(\beta)$ is known. The non-isolated equation can be rewritten as

$$y_i - \lambda \tilde{y}_i(\beta) = \delta x_i' \gamma + e_i.$$

The isolated equation is

$$y_i = x_i' \gamma + \varepsilon_i.$$

Thus, conditional on (λ, β, δ) , both equations are linear in the same unknown coefficient vector γ . The estimator can therefore solve for the best γ directly, and the numerical search only needs to vary the lower-dimensional vector (λ, β, δ) .

The concentrated estimator combines two steps:

1. an outer nonlinear search over (λ, β, δ) ;
2. an inner linear estimation of γ for each candidate value of (λ, β, δ) .

The big advantage of this approach is that for linear models GMM has a known closed-form solution. The inner step can therefore be solved quickly and exactly, without any numerical search. This makes computation faster and more stable while leaving the model and moment conditions unchanged.

Appendix: Concentrated OLS γ

This appendix derives the closed-form expression for γ used inside the GMM objective. The key observation is that, conditional on (λ, β, δ) , the moment conditions are linear in γ . Therefore, rather than numerically optimizing over γ , we can solve for its GMM-optimal value analytically.

Let I denote the sample of isolated individuals and N the sample of non-isolated individuals. After residualizing outcomes and covariates with respect to the relevant fixed effects, the isolated outcome equation is

$$y_I = X_I \gamma + \varepsilon_I,$$

while the non-isolated outcome equation is

$$y_N = \delta X_N \gamma + \lambda S_N(\beta) + e_N,$$

where $S_N(\beta)$ is the residualized peer norm evaluated at β . For the non-isolated observations, define the instrument matrix

$$Z_N(\beta) = \begin{bmatrix} X_N & \hat{S}_N(\beta) & \hat{D}_N(\beta) \end{bmatrix},$$

where $\hat{S}_N(\beta)$ is the residualized predicted peer norm and $\hat{D}_N(\beta)$ is its derivative with respect to β .

The isolated moment condition is

$$m_I(\gamma) = \frac{1}{N_I} X_I' (y_I - X_I \gamma).$$

This can be written as

$$m_I(\gamma) = a_I - B_I \gamma, \quad \text{where} \quad a_I = \frac{1}{N_I} X_I' y_I, \quad B_I = \frac{1}{N_I} X_I' X_I.$$

Similarly, the non-isolated moment condition is

$$m_N(\gamma) = \frac{1}{N_N} Z_N' (y_N - \lambda S_N(\beta) - \delta X_N \gamma).$$

This can be written as

$$m_N(\gamma) = a_N - B_N \gamma, \quad \text{where} \quad a_N = \frac{1}{N_N} Z_N' (y_N - \lambda S_N(\beta)), \quad B_N = \delta \frac{1}{N_N} Z_N' X_N.$$

For fixed (λ, β, δ) , the GMM objective as a function of γ is

$$Q(\gamma) = m_I(\gamma)' W_I m_I(\gamma) + m_N(\gamma)' W_N m_N(\gamma).$$

Substituting the linear representations of the two moments gives

$$Q(\gamma) = (a_I - B_I \gamma)' W_I (a_I - B_I \gamma) + (a_N - B_N \gamma)' W_N (a_N - B_N \gamma).$$

Taking the first-order condition with respect to γ gives

$$-2B_I' W_I (a_I - B_I \gamma) - 2B_N' W_N (a_N - B_N \gamma) = 0.$$

Rearranging,

$$B_I'W_I B_I \gamma + B_N'W_N B_N \gamma = B_I'W_I a_I + B_N'W_N a_N.$$

Therefore, the GMM-optimal value of γ , conditional on (λ, β, δ) , is

$$\hat{\gamma}(\lambda, \beta, \delta) = (B_I'W_I B_I + B_N'W_N B_N)^{-1} (B_I'W_I a_I + B_N'W_N a_N).$$

In implementation, the matrices are

$$B_I = \frac{1}{N_I} X_I' X_I, \quad a_I = \frac{1}{N_I} X_I' y_I,$$

and

$$B_N = \delta \frac{1}{N_N} Z_N' X_N, \quad a_N = \frac{1}{N_N} Z_N' (y_N - \lambda S_N(\beta)).$$

Thus the code constructs

$$A = B_I'W_I B_I + B_N'W_N B_N,$$

and

$$b = B_I'W_I a_I + B_N'W_N a_N,$$

then solves

$$\hat{\gamma} = A^{-1}b.$$